

IES E-CAMPUS: STUDENT EXPERIENCE & PORTAL MANUAL

Executive Presentation Deck & Operational Demonstration Guide

Document Purpose: Presentation Slide Deck for Evaluators, Faculty Mentors & Academic Management

Target Audience: College Leadership, Evaluators & Students

Institution: IES College of Technology / IES University, Bhopal (RGPV & AICTE Approved)

Version: 2.4.0 (Enterprise Academic Edition)

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[Slide 1] Executive Overview & The Paperless Campus Vision

1. The Real-College Problem

In typical engineering colleges, students lose valuable academic hours standing in physical lines:

- Waiting 3 to 7 days for paper Bonafide certificates needed for National Scholarship Portal (NSP) or MPTAAS.
- Finding out about attendance shortages only at the end of the semester when detention lists are pinned to bulletin boards.
- Missing classroom changes or substitute lectures due to stagnant paper timetables.
- Joining unmoderated WhatsApp groups where private mobile numbers are leaked and academic doubts get lost in spam.

2. The IES E-Campus Solution

A unified, paperless, mobile-first ecosystem connecting students, professors, and administration in real time with bank-grade security, zero server bills, and digital seal attestation.

STUDENT LIFE ECOSYSTEM		
ACADEMIC OS • Live Classes • Daily Timetable • 75% Attendance	ADMINISTRATIVE • PVC ID Card • Bonafide PDF • Sem Registration	COMMUNITY & CHAT • Batch Doubt Groups • Faculty 1-on-1 Q&A • Anti-Tamper Security

Presenter Talking Points

"Respected evaluators, today we present IES E-Campus. We did not build a generic demo; we built a functional, production-ready operating system that solves the everyday pain points of engineering students—from attendance detention fears to instant scholarship bonafide approvals."

 [SCREENSHOT PLACEHOLDER: Student Portal Hero Splash & Welcome Presentation]

[Slide 2] Student Onboarding, Two-Tier Verification & Security Gate

1. Step-by-Step Working Mechanism

- **Step 1 (Input):** Student taps "Student Portal" and enters:
 - Full Legal Name (as per 10th mark sheet)
 - RGPV Enrollment Number (e.g., 0103CS221045) & University Roll Number (220101)
 - Department (CSE), Current Year (3rd Year), Semester (5th Sem), Section (Section A)
 - College Email & Password
- **Step 2 (Processing):**
 - Database writes user account with `isApproved: false`.
 - App displays a secure **"Pending Faculty Verification"** lock screen.
 - An alert notification is dispatched to the faculty dashboard.
- **Step 3 (Output):**
 - Professor verifies roll number against class roster and taps **"Approve"**.
 - Student app unlocks automatically in real time without requiring a restart.

STUDENT REGISTRATION FORM	
Full Name:	[Rahul Sharma]
Enrollment No:	[0103CS221045]
Department:	[Computer Science & Engineering ▼]
Year & Semester:	[3rd Year (5th Sem) ▼]
Section:	[Section A ▼]
Email:	[rahul.sharma@iesbpl.ac.in]
Password:	[.....]
[ Create Account & Request Faculty Verification]	

Presenter Talking Points

"Security is our foundation. Anyone can download an APK, but no outsider can access internal campus lectures or batch discussion groups. Every student account must be authenticated by an active professor before access is granted."

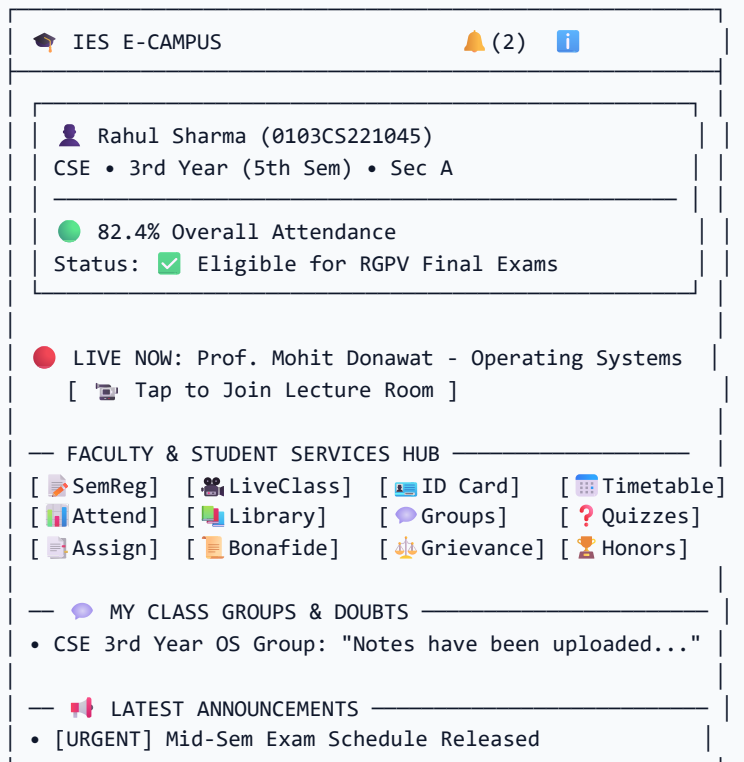
 [SCREENSHOT PLACEHOLDER: Student Signup Form & Pending Approval Security Screen]

[Slide 3] Student OS Home Dashboard & Real-Time Pulse

1. Exact Screen Architecture

1. **Pinned Header:** Official IES College crest, live notification bell counter, developer info.
2. **Hero Student Identity Card:**

3. Live Circular Attendance Percentage Meter.
4. Dynamic Status Chip: Green (ELIGIBLE FOR EXAMS), Amber (CONDONATION ZONE), or Red (DETENTION WARNING).
5. **Active Live Lecture Banner:**
6. Appears dynamically with a pulsing red badge whenever faculty is broadcasting.
7. **4-Column Quick Services Hub:**
8. 14 tools laid out for one-thumb navigation.
9. **Class Discussion Groups Box:**
10. Shows active batch channel with coordinator name and latest message snippet.
11. **Campus Announcements Feed:**
12. Live stream of priority circulars.



Presenter Talking Points

"This dashboard is the student's command center. The moment a student unlocks their phone, they instantly see their exam eligibility percentage, whether a class is live, today's schedule, and class doubts in under 3 seconds."

 [SCREENSHOT PLACEHOLDER: Student Dashboard Home Screen with Attendance Ring]

[Slide 4] Hanging Lanyard PVC Identity Card System

1. Step-by-Step Working Mechanism

- **Input:** Pulls student credentials and attested profile photo automatically.
- **Processing:**
- **Front Assembly:** Draws metallic oval lanyard punch-hole band, institution header, golden session ribbon (2024 - 2026), passport photo with holographic security border, double-ring red circular seal, registrar signature, and barcode.
- **Back Assembly:** Draws Father's Name, emergency contact, residential address, return address, and 5-point institutional discipline rules.
- **Export Formats:**

- **Gallery PNG:** Flutter `RepaintBoundary` rasterizes the widget matrix at **300+ DPI** directly into phone storage.
- **Print-Ready A4 PVC PDF:** Renders Front and Back side-by-side with alignment markers for direct feeding into PVC card printing machines.

VERTICAL LANYARD ID CARD

<-- Metallic Lanyard
Punch Hole

★ IES COLLEGE OF TECHNOLOGY, BHOPAL ★

SESSION 2024 - 2025

PHOTO
[ATTESTED]

RAHUL SHARMA

COMPUTER SCIENCE & ENGG

Enrollment No : 0103CS221045	Blood Group : B+
Roll Number : 220101	Semester : 5th
Batch Year : 3rd Year	Section : A

(★ RED COLLEGE SEAL ★)
(Registrar Signature)

Presenter Talking Points

"Unlike horizontal demo cards, our ID card is a vertical hanging lanyard card with an authentic punch slot, university crimson seal, and barcode. It can be printed directly on PVC plastic for official campus gate access."

[SCREENSHOT PLACEHOLDER: Front and Back View of Lanyard ID Card with Red Seal Stamp]

[Slide 5] Live Interactive Lecture Room (PhysicsWallah / Unacademy Style)

1. Step-by-Step Working Mechanism

- **Trigger:** When professor starts broadcasting, student sees the glowing red alert banner.
- **Action:** Student taps "Join Lecture".
- **Behind-The-Scenes Processing:**
 - Embedded HD video player mounts the live stream (YouTube Live / WebRTC stream).
 - Student UID is logged to mark real-time attendance for that lecture period.
 - Real-time Q&A stream connects to the lecture doubts channel.
- **Live Doubts Chat:**
 - Student types a question.
 - Faculty responses are highlighted in golden cards with verified **FACULTY** tags.
- **Handouts:** Lecture slides (PPT/PDF) are available for immediate download directly below the video player.

LIVE: Operating Systems - Unit 3 Memory Management

[LIVE VIDEO PLAYER]

Prof. Mohit Donawat (HOD)

LIVE LECTURE CHAT:

- Amit: Sir, can you explain page replacement once?
- 🇮🇳 Prof. Mohit: Yes Amit, we will solve LRU now.
- Priya: Notes PDF download button working perfectly.

[Type your doubt to Professor...]

[Send]

👤 Presenter Talking Points

"Online learning should not be a static Zoom link. Our live lecture experience integrates automated attendance upon joining, embedded video streaming, real-time doubts chat, and direct slide handout downloads in one unified view."

🖼️ [SCREENSHOT PLACEHOLDER: Live Class Player with Student Chat and Slide Handout Tabs]

[Slide 6] Semester Registration & Course Enrollment Portal (RGPV Standard)

1. Step-by-Step Working Mechanism

- **Input:**
 - Bio-data auto-populates from profile (Name, Enrollment, Branch, Father's Name).
 - Student enters: Target Semester (5th Sem), Academic Session (2024-2025), Previous Semester SGPA (8.42), Cumulative CGPA (8.10).
 - Backlog Toggle: If Yes, inputs subject codes.
 - Extra Achievements: Hackathons, NPTEL certificates, sports.
 - Elective Course Choices & Fee Payment Reference Number.
 - Mandatory Undertaking: Checkbox agreeing to 75% attendance and anti-ragging compliance.
- **Processing:**
 - Document saved in semester_registrations with status PENDING .
 - Dispatched to HOD review queue.
 - Upon HOD approval, digital HOD signature and crimson red seal stamp are applied.
- **Output:** Official print-ready A4 PDF formatted across Part A to Part E with unique electronic reference code (IES/REG/2024/0103CS221045).

SEMESTER REGISTRATION & ENROLLMENT FORM

Applying For	: [5th Semester ▼]
Previous SGPA	: [8.42]
Cumulative CGPA	: [8.10]
Any Backlogs?	: [NO Toggle]
Achievements	: [Smart India Hackathon Winner 2024]
Elective Choice	: [Advance Database Management Sys ▼]
Fee Status	: [PAID - Ref: IES/FEE/2024/99128]

[X] I declare that I will maintain min 75% attendance.

[📄 Submit Form & Download Official Attested PDF]

👤 Presenter Talking Points

"Every semester, thousands of students wait in line to submit paper registration forms. Our system generates an authentic, AICTE/RGPV compliant five-part registration document with fee verification and HOD attestation in seconds."



[Slide 7] Instant Official Bonafide Certificate Generator

1. Step-by-Step Working Mechanism

- **Step 1 (Select Purpose):** Student chooses from templates (NSP Scholarship, MPTAAS, Education Loan, Bus Pass, Passport) or enters a custom reason.
- **Step 2 (Attestation Assembly):**
 - System generates unique serial: `IES/BONA/2024/[SERIAL]`.
 - Institutional phrasing: "This is to certify that Rahul Sharma, Son of Shri Ramesh Sharma..."
 - Embeds attested student passport photo.
 - Double-ring crimson red official college seal stamp.
 - Authorized Registrar / Principal signature block.
- **Dynamic Security QR Code:** Carries encrypted JSON string containing candidate UID, enrollment number, branch, and issue date.
- **Step 3 (Download & Share):** Generates high-res A4 PDF instantly with authoritative legal validity guarantee banner.

BONAFIDE CERTIFICATE (अध्ययनरत प्रमाण पत्र)	
Reference No: IES/BONA/2024/7819	Date: Today
This is to certify that Mr. RAHUL SHARMA Son of Shri RAMESH SHARMA, Enrollment: 0103CS221045 is a bonafide regular student of 3rd Year (5th Sem) B.Tech in Computer Science & Engineering at this Institute during the academic session 2024-2025.	
Purpose: National Scholarship Portal (NSP) Application	
(★ RED COLLEGE SEAL ★) [QR CODE]	(Registrar Signature) IES University, Bhopal

Presenter Talking Points

"Instead of waiting a week for a clerk's signature, our students generate a digitally attested Bonafide certificate in 5 seconds. The QR code allows any government scholarship officer or bank to scan and verify authenticity immediately."



[Slide 8] Timetable Tracker & Substitute Class Radar

1. Step-by-Step Working Mechanism

- **Input:** Automatically queries timetable using student's Branch, Year, Semester, and Section.
- **Processing:**
 - Checks current weekday (Monday = 1).
 - Fetches the active period list from Firestore/local memory cache.
 - Checks if a day-specific **Substitute Lecture** has been scheduled by faculty for today.
- **Display:**
 - Displays periods 1 to 8 with time ranges, subjects, professor names, and classroom/lab room numbers.

- If a substitute professor is covering a class, highlights a yellow [⚡ SUBSTITUTE] badge showing the substitute teacher and temporary room.
- Includes a photo viewer tab to view the physical notice board schedule with pinch-to-zoom.

📅 TODAY'S SCHEDULE - MONDAY (6 Periods)	
[🟢 Mon]	[Tue] [Wed] [Thu] [Fri] [Sat]
09:00 AM - 10:00 AM • Period 1 Operating Systems (CS-501) 🧑‍🏫 Prof. Mohit Donawat • 📍 Room 304	
10:00 AM - 11:00 AM • Period 2 [⚡ SUBSTITUTE] Database Management Systems (CS-502) 🧑‍🏫 Prof. Sharma (Covering) • 📍 Computer Lab 2	

🗣️ Presenter Talking Points

"Stagnant paper schedules create confusion when teachers take leave. Our timetable engine supports dual scheduling: a recurring weekly master timetable and real-time substitute updates for today's classes."

🖼️ [SCREENSHOT PLACEHOLDER: Student Daily Timetable Screen with Period Cards]

[Slide 9] Smart Attendance Meter & 75% Prediction Engine

1. Mathematical Mechanics & Formulas

- **Live Attendance Percentage:**

$$\text{Percentage} = \left(\frac{\text{Total Attended}}{\text{Total Held}} \right) \times 100\%$$
- **Color Coding Logic:**
 - 🟢 **Green ($\geq 75\%$):** Exam Eligible.
 - 🟡 **Amber ($65\% \leq \text{Percentage} < 74.9\%$):** Condonation / Warning Zone.
 - 🔴 **Red ($< 65\%$):** Detention Zone.
- **Smart Prediction Formula (When Below 75%):**

$$\text{Classes Needed} = \left\lceil \frac{0.75 \times \text{Total Held} - \text{Total Attended}}{0.25} \right\rceil$$

Example: If a student attended 20 out of 30 classes (66.7%), the system computes that they must attend the next **10 classes consecutively** to cross 75.0% .
- **Safe Margin Calculation (When Above 75%):**
 Displays exactly how many classes can be safely skipped while staying at or above 75%.

🗣️ Presenter Talking Points

"Under RGPV regulations, attendance below 75% results in exam detention. Our system doesn't just show numbers; its prediction engine tells students exactly how many classes they must attend consecutively to reach safety."

🖼️ [SCREENSHOT PLACEHOLDER: Student Attendance Progress Bar and Subject-Wise Breakdown]

[Slide 10] Digital Library & Academic Handouts Repository

1. Step-by-Step Working Mechanism

- **Branch & Semester Filtering:** Automatically filters resources matching the student's active subjects.
- **4 Document Categories:**
 -  Unit-wise handwritten notes & PPT lecture slides.
 -  Previous Years Question Papers (RGPV PYQs with solutions).
 -  Laboratory manuals & practical code exercises.
 -  Reference text PDFs.
- **In-App PDF Viewer:** Open, pinch-zoom, and study documents directly inside the app.
- **Offline Download:** Save documents to phone storage for study during daily bus commutes.

Presenter Talking Points

"Students no longer need to spend money on photocopies or search spam-filled websites for previous year question papers. Everything curated by their own department faculty is available in one tap."

 **[SCREENSHOT PLACEHOLDER: Digital Library Screen with Subject Folders and PDF Viewer]**

[Slide 11] Interactive Timed Quizzes & Exam Preparation

1. Step-by-Step Working Mechanism

- **Test Engine:**
 - Timed MCQ tests designed by department faculty for Mid-Sem and GATE preparation.
 - Live countdown clock displaying remaining minutes and seconds.
 - Question status tracker: Answered (Green), Unanswered (Grey), Marked for Review (Purple).
- **Instant Result & Scorecard:**
 - Submit test $\rightarrow$ Automated instant evaluation.
 - Scorecard displays total score, accuracy percentage, time taken, and correct answer explanations.
 - Batch leaderboard ranking.

Presenter Talking Points

"Online assessments provide immediate feedback. Students see their score, review detailed explanations, and check their batch rank the moment they submit."

 **[SCREENSHOT PLACEHOLDER: Quiz Question Screen with Countdown Timer and Result Analysis]**

[Slide 12] Assignment Submission & 6-Month Review Archive

1. Step-by-Step Working Mechanism

- **Assignment Notification:** Live push alert when a professor dispatches an assignment problem sheet with deadline and max marks.
- **Submission Upload:**
 - Student takes photos of handwritten sheets or uploads a typed PDF.
 - System generates an automatic timestamp verification receipt.
- **6-Month Review Archive:**
 - Retains all graded assignments, professor marks, and personalized feedback for half a year for academic audit compliance.

Presenter Talking Points

"No more lost paper assignments. Submissions carry digital timestamp receipts, and students can review all past marks and feedback across the entire semester."

 **[SCREENSHOT PLACEHOLDER: Assignment Details Screen with File Upload and Grade Receipt]**

[Slide 13] Anti-Tamper Class Discussion Groups & Doubt Clearing Hub

1. Step-by-Step Working Mechanism

- **Targeted Batch Groups:** Students only see groups matching their specific Branch, Year, Semester, and Section.
- **Canonical Branch Matching Engine:**
 - Automatically matches abbreviations: CSE $\rightarrow$ Computer Science & Engineering, ME $\rightarrow$ Mechanical Engineering, Civil $\rightarrow$ Civil Engineering.
- **Instant Optimistic Messaging:**
 - Message bubble renders in 1 millisecond on screen while syncing to Firestore in the background.
- **Quick Academic Doubt Chips:**
 - 1-tap prompts: " ? Ask a Question", " 📅 When is submission?", " 📖 Notes Needed", " 🗣️ Attendance Query".
- **Anti-Tamper & Security Rules:**
 - **No Deletion:** Messages cannot be deleted once posted, maintaining academic decorum.
 - **60-Second Edit Window:** Typos can only be corrected within 1 minute with an [edited] tag.
 - **Verified Badges:** Professors have golden 🏆 FACULTY cards, preventing student impersonation.

👤 Presenter Talking Points

"WhatsApp groups lead to privacy issues and message deletion to hide indiscipline. Our built-in discussion hub keeps mobile numbers private, prevents message deletion, and provides verified badges for professors."

📸 [SCREENSHOT PLACEHOLDER: Group Chat Room with Quick Chips and Faculty Golden Bubble]

[Slide 14] Statutory Grievance Redressal & Anti-Ragging Cell

1. Step-by-Step Working Mechanism

- **Confidential & Anonymous Filing:** Students can report issues with their identity or completely anonymously under UGC guidelines.
- **Categories:** Academics, Infrastructure, Hostel, Transportation, Harassment, Ragging.
- **Live Ticket Status Tracking:**
 - SUBMITTED $\rightarrow$ UNDER REVIEW $\rightarrow$ ACTION TAKEN $\rightarrow$ RESOLVED.
- **Administrative Action Taken Report (ATR):** Committee records corrective actions taken, creating an auditable log for annual AICTE inspections.

👤 Presenter Talking Points

"Student welfare is paramount. Under UGC and AICTE regulations, our grievance desk gives students a safe channel to raise concerns anonymously while tracking resolution progress."

📸 [SCREENSHOT PLACEHOLDER: Grievance Submission Screen and Ticket Progress Tracker]

[Slide 15] Circulars, Announcements & Hall of Fame

1. Step-by-Step Working Mechanism

- **Priority Circulars:**
 - **🔴 URGENT:** High-priority alert banner (e.g., Holiday declared due to heavy rain, Mid-Sem date sheet release).
 - **🟡 NORMAL:** General club and seminar updates.
- In-app PDF viewer for official university gazette orders.

- **Campus Hall of Fame:**
- Highlights top placement packages (e.g. *12 LPA, 18 LPA offers*), university rank holders, and Smart India Hackathon (SIH) winners to inspire junior batches.

Presenter Talking Points

"Important circulars are delivered directly to student devices in seconds, while our Hall of Fame celebrates student achievements to foster academic ambition across campus."

 **[SCREENSHOT PLACEHOLDER: Campus Announcements Feed and Hall of Fame Recognition Wall]**

End of Student Presentation Deck • Ready for Projection & Evaluation